WSDB Access from Python

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General query tool [edit]

The database can be accessed using the sqlutilpy module https://github.com/segasai/sqlutilpy You can install it using

```
pip install sqlutilpy
```

If you have installed sqlutil then you can access the database using a very simple syntax, which directly outputs query results into numpy arrays:

```
ra,dec = sqlutilpy.get('select ra, dec from sdssdr9.phototag limit 10',
                    db='wsdb',host='wsdb.ast.cam.ac.uk', user='your_login', password='your_password')
```

(Note: if you don't want to type the password all the time, you can save it in .pgpass file see here WSDB_Introduction_into_PSQL_client If you use the environmental variable PGHOST PGDATABASE PGUSER, you won't need to use specify db/user/host options either.

An important thing you could do with sqlutil is also match with your data stored as numpy arrays. I.e here is an example of a complication spatial x-match

```
yw1,yw2,yw3,yj,yh,yk = sqlutilpy.local_join("""
 select w1mpro,w2mpro,w3mpro,j_m_2mass,h_m_2mass,k_m_2mass from mytable
     as m left join lateral (select * from allwise.main as  s where
             q3c_join(m.ra, m.dec,s.ra,s.dec,1./3600)
             and ext_flg=0 and cc_flags like '00%'
     order by q3c_dist(m.ra,m.dec,s.ra,s.dec) asc limit 1)
     as tt on  true  order by xid """,'mytable',
     (ra,dec,np.arange(len(dec))),('ra','dec','xid'),
         host='wsdb.ast.cam.ac.uk',
     preamb='set enable_seqscan to off; set enable_mergejoin to off; set enable_hashjoin to off;', db='wsdb')
```

Here we assume that you have ra,dec numpy arrays

You can also use sqlutilpy to upload data

Easy Crossmatching by position [edit]

If you need to crossmatch your data with some table in WSDB you can use the crossmatcher module https://github.com/segasai/astrolibpy/blob/master/my_utils/crossmatcher.py

I.e. this commands retrieves the g,r magnitude for your catalog of ra,decs using 2" matching radius

```
gmag,rmag= crossmatcher.doit('sdssdr9.phototag', ra,dec,'psfmag_g,psfmag_r', rad=2.,
          host='wsdb.ast.cam.ac.uk', db='wsdb')
```

You can also crossmatch by key. I.e. if you have a list of source_ids and want to fetch the measurements for that list:

```
rv,= crossmatcher.doit_by_key(
   "gaia_dr3.gaia_source",
   source_id_arr,
   'radial_velocity',
   key_col='source_id',
   host="wsdb.ast.cam.ac.uk",
   db="wsdb")
```